

Exhibit 7

NEWTON's TELECOM DICTIONARY

27th Expanded and Updated Edition

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New York

NEWTON's TELECOM DICTIONARY

27th Expanded and Updated Edition

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signals. The timing elements at the sending and receiving terminal define where one character ends and another begins. There are no start or stop elements in this form of transmission. For a more detailed explanation, see BSC.

binary to decimal conversion Conversion from base 2 to base 10. See Binary.

binary transfer See Binary File Transfer and MIME.

BIND Berkeley Internet Name Daemon. BIND is the software that allows us to type site names like *www.yahoo.com* instead of a string of numbers. BIND is the implementation of a DNS server originally developed and distributed by the University of California at Berkeley. Many Internet Hosts run BIND, and it is the ancestor of many commercial BIND implementations.

bind A request to activate a session between two logical units (LUs). See BIND.

binder A way to separate groups of 25 pairs in a twisted-pair cable with more than 25. Color plastic ribbon binds separate each group of 25. The first group (1-25) is white/blue, the second (26-50) is white/orange, the third (51-75) is white/green, the fourth (76-100) is white/brown, etc. Binders are helically applied colored thread, yarn, or plastic ribbon. They're used to confine, separate and identify groups of fibers or wires in a cable. Binders are usually used for holding assembled cable components in place.

binder group A typical phone line—business or residence, analog or digital—is a pair of wires. When the phone company installs cabling along the street—up in the air or underground—it installs a cable containing many pairs of wires. The pairs are color coded so you can find which is which. Often there are so many of them you have to do something better—like take a bunch of cable pairs and wind something (or bind something) around them. That's called a binder group. The binder is a spirally wound colored thread or plastic ribbon used to separate and identify cable pairs by means of color coding. The groups are composed of insulated twisted copper pairs that are also twisted within each binder. Typically, they are wrapped in 25-pair bundles. Normal telephone color-coding provides for only 25 pairs of wire, so binder groups allow multiple pairs of the same color wire to be in one cable. A 50-pair cable has blue and orange binder groups; a 75-pair cable has blue, orange, and green groups. Since several wire pairs have the same color markings, installers must be careful when stripping cable insulation so they do not destroy the binder threads. For example, pairs 1-25 might be in one binder group and pairs 26-50 in another. In xDSL, one often hears discussions of signal interference between adjacent pairs within a binder group. The best of all worlds is to keep a data pair separated from another data pair by assigning it to an adjacent binder group. If the data pairs are too close to each other they create what telephone companies call "disturbances" (i.e. crosstalk).

bindery A Novell NetWare database containing definitions for entities such as users, groups, and workgroups. The bindery contains three components: objects, properties, and property data sets. Objects represent any physical or logical entity, including users, user groups, file servers, print servers, or any other entity given a name. Properties are the characteristics of each bindery object, including passwords, account restrictions, account balances, internetwork addresses, list of authorized clients, workgroups, and group members. Property data sets are values assigned to entities' bindery properties.

binding A Windows definition. Binding is a process that establishes the communication channel between a protocol driver and a network adapter driver.

binding post 1. BP. A screw with a small nut. You take your phone wires and join them together on a binding post by wrapping them together around the screw and then tightening the nut on them. You'll find binding posts on huge wall-mounted things called terminal boxes. We have several in the basement of the building in Manhattan, New York, in which I live. Binding posts are numbered on the terminal box and those numbers are entered into the phone company's cable management system. That way, when the technician comes to check out a trouble report, he can quickly find the offending pairs. These days, other more modern termination devices (such as Krone Boxes, 110-connecting blocks, and even 66-blocks) are replacing binding posts. Our building's binding posts are probably 60 years old. In telephone parlance, the binding post is neither a rate demarcation point or a network interface. The binding

post is part of the telephone company's plant and cannot be accessed by the telephone subscriber. The only exception I can think of is the wholesale (CLEC) scenario. The issue of who would have access to the binding post would depend on the agreement signed by the ILEC and CLEC. The reason for limiting access to binding posts is simple: If retail customers have access to binding posts for do-it-yourself tinkering, they could accidentally or maliciously cause trouble. In a shared scenario (for example, in an apartment building), a subscriber could connect his inside wire to his neighbor's binding post and listen to his neighbor's phone calls and make long-distance calls on his neighbor's dime. Or he could clumsily damage LEC facilities. See also House Box.

2. A post that witches were attached to just before they were burned. Rude switchboard operators and ACDs with long queues should suffer this fate.

BinHex When sending files which aren't plain ASCII across a network—dial-up, leased line or the Internet—you basically have two options. First, you can attach them as a binary file (i.e. non-ASCII file). Or second, you can encode them into ASCII characters and send the file as part of your message. The first method is preferable. But you can typically only send binary files from one account to another on the same network, or between two networks that have agreed between themselves to a method of transferring files. But that's a rarity. And it certainly doesn't work in and around the Internet. Thus something called MIME was created. MIME stands for Multipurpose Internet Mail Extensions. It is the name for encoding binary files into ASCII characters for transfer across the Internet and on-line services. Under MIME, there are a number of methods of encoding binary files, one of which is called BinHex, which is a popular encoding algorithm that uses Run-Length Encoding (RLE). To send a binary file, you encode it using BinHex encoding software. You then include the ASCII encoded file in your message. The recipient of your message then decodes the ASCII using BinHex decoding software, which may or may not be built into his electronic mail or browser package.

binomial distribution The binomial is a two-parameter distribution, the parameters being *n*, the number of trials, and *p*, the probability of a particular outcome of a single trial.

bio break Meeting-speak for "bathroom break." "Let's take a short bio break before moving on to action plans."

bio-surveillance Monitoring hospital admissions to detect increases in similar symptoms that could indicate a biological attack.

biobology The art of scrutinizing and interpreting ambiguous, lo-res imagery. Practiced by spy agencies and, more recently, neuroscientists studying brain scans.

biogorrrhea Pointless or excessive filler in a weblog. As one Net diarist wrote: "I was accused recently of biogorrrhea. Well, yes, that's fair. But see, here's the thing. I'm leaving town for a month..." Courtesy, Gareth Branwyn, *Wired Magazine*.

biohacker A hobbyist who tinkers with DNA and other aspects of genetics. Also: bio-hacker. Some speculate a "biohacker"—the equivalent of a computer hacker seeking thrills rather than impact—could be behind the anthrax letters. "Hacker" is usually considered a bad word.

bioinformatics Bioinformatics is the process of using computer systems for the purposes of acquisition, storage and analysis of biological data. Computer databases and algorithms are used to speed up biological research and to characterize the molecular components of living things. Bioinformatics is notably being used in the Human Genome Project, the effort to identify the 80,000 genes in human DNA. Bioinformatics includes a wide spectrum of technologies including computer architectures, knowledge management, networking and collaboration tools as well as traditional life-science equipment needed to handle biological samples.

biomass addition New growth of vegetation. See vegetation management.

biometrics Biometrics is the study of unique and measurable physical (or biological) characteristics such as fingerprints, retinal pattern, or handwriting, biometric authentication devices capture, encrypt and use these unique and measurable characteristics as the basis for confirming identity and determining whether to grant or deny access to physical or logical assets. When used for au-

thentication, personal biometrics are used much like a door key, however that key cannot be duplicated or lost. The basic principles of biometric verification have been understood and practiced for thousands of years. References occur as early as ancient Egypt, where individuals were identified via unique "biometrics" such as scars or a combination of measured features such as complexion, eye color, height and so on. Biometrics as we know it today really began in the late nineteenth century when criminology researchers attempted to tie physical features and characteristics with criminal tendencies. The results were not conclusive but the idea of measuring individual physical characteristics became accepted, leading the way for fingerprinting to become the international methodology among police forces for identity verification. Today it is taken for granted that fingerprints (and foot and handprints) accurately identify not just criminals, but newborns, minors (as a means of thwarting abduction), military and police personnel, and a host of others. Once technology simplified the collection, codification and encryption of biometric data, the stage has become set for biometrics to become the security standard of the electronic world, replacing passwords, bits of paper, etc. See also biometric access control.

biometric access control Any means of controlling access through human measurements, such as fingerprinting, handwriting analysis, retinal scanning, hand geometry and voiceprinting. See Biometrics.

biometric device A device used in authenticating access to a system. A biometric device authenticates a user by measuring some hard-to-forge physical characteristic, such as a fingerprint or retinal scan. You see a lot of biometric devices in Hollywood thrillers, including *Mission Impossible*, a movie which was released in the summer of 1996.

bionic code A problem-solving routine for human behavior as it is exercised in the realm of networks and cyberspace. The first bionic codes were developed by Ebon Fisher based on a series of his theatrical experiments involving communication systems amongst audience members. Fisher's bionic codes have been formalized as a series of diagrams and statements which "float" in the infosphere in a variety of media.

BIOS Basic Input/Output System of desktop computers. The BIOS contains the buffers for sending information from a program to the actual hardware device the information should go to. Every personal computer has a system basic input/output system, or BIOS, which is what takes control of your computer the moment you turn it on. The screen that you first see when you turn on your computer is called the power on self test screen, better known as the POST screen. If you purchased your computer from one of the major computer manufacturers, this screen is often hidden by the manufacturer's logo. To get rid of this logo from the screen, press the ESC button on your keyboard; you'll then see what is going on in the background. At this stage in the system boot, the BIOS is probing the hardware to test the system memory and other device connections. Once the POST is completed, the BIOS proceeds to look for a device to boot from. Once it finds your hard drive, it will begin to load Windows. The BIOS also acts as a main system component control panel, where low-level settings for all of your hardware devices are made. The device boot order, port addresses, and feature settings such as plug and play are all found in the BIOS setup screens. For example, if you want to change the order of the drives that your computer checks to boot from, then you will want to modify the device boot order. I have to modify this setting almost every time that I install Windows because I want my computer to boot off the CD-ROM to launch the Windows XP setup application instead booting off of the operating system on my hard drive. BIOSs on each and every PC may be made by different companies or accessed in different ways. Nevertheless, the most common way to access the setup screen is to press F2 or the Delete key when the POST screen is displayed. Some computers even tell you which key to push to enter setup, as my laptop does. If your PC doesn't allow you to access the setup screen in this way, consult your computer documentation or contact your computer manufacturer for instructions. See also extensible firmware interface.

BIOS Enumerator A Windows term. Bios enumerator is responsible, in a Plug and Play system, for identifying all hardware devices on the motherboard of the computer. The BIOS supports an API that allows all Plug and Play computers to be queried in a common manner. See BIOS.

biosensors Devices such as fingerprint readers and signature recognition systems.

BIP 1. Billing Interconnection Percentage. A calculation of who owns how much of a route when there are multiple providers between two points. Used to allocate revenue in MPB arrangements.

2. See Broadband Initiatives Program.

BIP-8 Bit Interleaved Parity 8. A method of error monitoring used in SONET optical fiber transmission systems. A SONET frame of data comprises a large number of bits organized into bytes, or 8-bit values. The bytes, of course, are interleaved into a byte stream. The BIP-8 method looks across all matching bit positions for those distinct bytes in a frame, and calculates parity, which should be even, rather than odd. If an odd parity is calculated at the receiving end, a bit error is indicated. The number of such errors in frames over a period of time constitutes the Bit Error Rate (BER). The specific period of time involved is known as a "sliding time window," as it varies according to the maximum detection time, which is sensitive to the transmission rate, or level of the Optical Carrier (e.g. OC-1, OC-3, OC-24, OC-48, OC-192). The higher the OC-N level, the higher the rate of transmission, and the shorter the sliding time window. In other words, the faster the rate of transmission, the more quickly the error detection process must take place in order to establish the BER and to correct the problem.

BIP-N Bit Interleaved Parity N. A method of error monitoring. With even parity, an N bit code is generated by the transmitting equipment over a specified portion of the signal in such a manner that the first bit of the code provides even parity over the first bit of all N-bit sequences within the specified portion, etc. Even parity is generated by setting the BIP-N bits so that there are an even number of 1s in each of all N-bit sequences including the BIP-N. See BIP-8 for a concrete example.

bipolar 1. The predominant signaling method used for digital transmission services, such as DDS and T-1. The signal carrying the binary value alternates between positive and negative. Zero and one values are represented by the signal amplitude at either polarity, while no-value "spaces" are at zero amplitude.

2. Bipolar Signal: A signal that can take on two polarities, of which neither is zero.

bipolar coding The T carrier line coding system that inverts the polarity of alternate "one" bits.

bipolar signal Any mechanism that converts digital data to digital signals on a wire through the use of two non-zero voltage levels—positive and negative—to encode binary 1s, and a zero-voltage level to encode binary 0s. The term "bipolar" literally means "two poles." The prefix "bi" in "bipolar" comes from a Latin word meaning "two." The term "bipolar" refers to the aforementioned fact that two different voltage polarities (positive and negative) are used to encode 1s; the first occurrence of a 1 in a bitstream is encoded as a positive voltage on the wire; the second occurrence of a 1 in the same bitstream is encoded as a negative voltage; the third occurrence of a 1 in the bitstream is encoded as a positive voltage; and so on. Bipolar encoding mechanisms have an advantage over unipolar encoding mechanisms in that the greater variability in voltage levels and polarities that is produced by the former makes it easier for the receiver to figure out where each bit begins and ends, especially when there is an unvarying sequence of voltage pulses (representing a string of identical bits). Multiple bipolar encoding techniques have been developed, including, but not limited to, alternate mark inversion, B8ZS, and HDB3. You can find explanations of these terms in this dictionary. Also see unipolar encoding.

bipolar violation The presence of two consecutive "one" bits of the same polarity on the T carrier line. See also Bipolar Coding.

Bipolar with 8 Zero Substitution See B8ZS.

bird A satellite.

bird dog device An electronic tracking device used by law enforcement to track the physical location of a subject.

birdie 1. A birdie is a lightweight device that you blow through underground cement pipes through which you want to pull cable. Here's how it typically works: First, you use a bore to make an underground hole. Then you fill that hole with hollow concrete cement pipes joined together to form one long un-